

# SSILP Math

Beginner

Analytic Trigonometry 1



# Analytic Trigonometry

*What you need to know*

- Trigonometric identities
- Addition and subtraction formulas
- Double angle and half angle formulas

*Refer to the textbook!*

Pre-calculus : Mathematics for Calculus  
By Stewart, Redlin, Watson (7<sup>th</sup> Ed)

**Chapter 7**

# Session outline

1. Trigonometric Identities

2. Addition and Subtraction  
formulas

3. Double-angle and Half-angle  
Formulas

# Learning objectives

- Understand and apply trigonometric identities, including addition, subtraction formulas
- Differentiate between equations and identities, and learn how to simplify trigonometric expressions using fundamental identities
- Utilize addition and subtraction formulas to find exact values of trigonometric expressions and simplify results
- Rewrite and simplify expressions involving inverse trigonometric functions in terms of variables

# 1. Trigonometric Identities

# Motivation

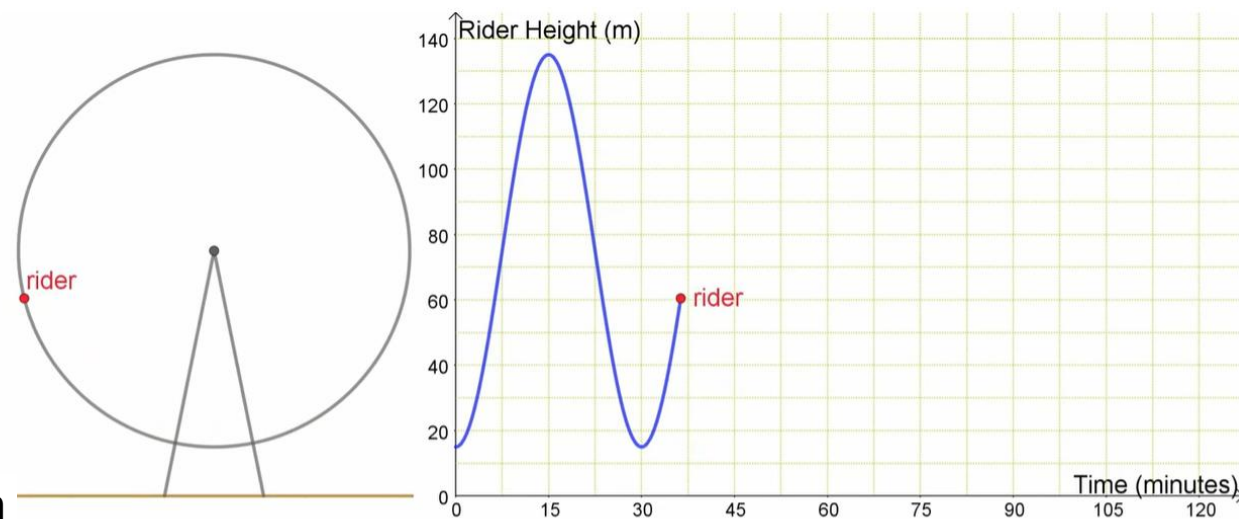
How can we **simplify** and **factor expressions**, and **solve equations** that involve trigonometric functions?

## Example:

Height of the rider on the London Eye Ferris Wheel

$$h(t) = -65 \cos\left(\frac{\pi}{15}t\right) + 70.$$

What is the length of time during which the rider is more than 100 meters above ground?



# Equation vs Identity

7

An **equation** is a statement that two mathematical expressions are equal

$$x + 2 = 5$$

not true for values of  $x$  other than 3.

An **identity** is an equation that is true for **all values of the variable(s).**

$$(x + 1)^2 = x^2 + 2x + 1$$

$$\sin^2 t + \cos^2 t = 1$$

# Quiz: Which of the following is **not** an identity?

(a)  $(x + y)^2 = x^2 + 2xy + y^2$

(b)  $x^2 + y^2 = 1$

(c)  $x(y + z) = xy + xz$

(d)  $\sin t + \cos t = 1$

# ACTIVITY



# Simplifying trigonometric expressions

SOLUTIONS

# Fundamental Trigonometric Identities

## FUNDAMENTAL TRIGONOMETRIC IDENTITIES

### Reciprocal Identities

$$\begin{aligned}\csc x &= \frac{1}{\sin x} & \sec x &= \frac{1}{\cos x} & \cot x &= \frac{1}{\tan x} \\ \tan x &= \frac{\sin x}{\cos x} & \cot x &= \frac{\cos x}{\sin x}\end{aligned}$$

### Pythagorean Identities

$$\sin^2 x + \cos^2 x = 1 \quad \tan^2 x + 1 = \sec^2 x \quad 1 + \cot^2 x = \csc^2 x$$

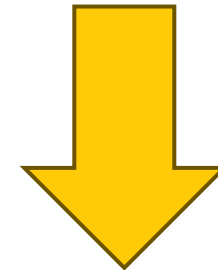
### Even-Odd Identities

$$\sin(-x) = -\sin x \quad \cos(-x) = \cos x \quad \tan(-x) = -\tan x$$

### Cofunction Identities

$$\begin{aligned}\sin\left(\frac{\pi}{2} - x\right) &= \cos x & \tan\left(\frac{\pi}{2} - x\right) &= \cot x & \sec\left(\frac{\pi}{2} - x\right) &= \csc x \\ \cos\left(\frac{\pi}{2} - x\right) &= \sin x & \cot\left(\frac{\pi}{2} - x\right) &= \tan x & \csc\left(\frac{\pi}{2} - x\right) &= \sec x\end{aligned}$$

Identities enable us to  
**write the  
same expression  
in  
different ways**



**Simplify  
complicated  
looking  
expressions**

# Simplifying a Trigonometric Expression

Simplify the expression  $\cos t + \tan t \sin t$ .

**SOLUTION** We start by rewriting the expression in terms of sine and cosine.

$$\begin{aligned}\cos t + \tan t \sin t &= \cos t + \left( \frac{\sin t}{\cos t} \right) \sin t && \text{Reciprocal identity} \\ &= \frac{\cos^2 t + \sin^2 t}{\cos t} && \text{Common denominator} \\ &= \frac{1}{\cos t} && \text{Pythagorean identity} \\ &= \sec t && \text{Reciprocal identity}\end{aligned}$$

# Simplifying by combining factors

Simplify the expression  $\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{1 + \sin \theta}$ .

**SOLUTION** We combine the fractions by using a common denominator.

$$\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{1 + \sin \theta} = \frac{\sin \theta (1 + \sin \theta) + \cos^2 \theta}{\cos \theta (1 + \sin \theta)} \quad \text{Common denominator}$$

$$= \frac{\sin \theta + \sin^2 \theta + \cos^2 \theta}{\cos \theta (1 + \sin \theta)} \quad \text{Distribute } \sin \theta$$

$$= \frac{\sin \theta + 1}{\cos \theta (1 + \sin \theta)} \quad \text{Pythagorean identity}$$

$$= \frac{1}{\cos \theta} = \sec \theta \quad \text{Cancel, and use reciprocal identity}$$

EXAMPLE

# Simplifying trigonometric expressions

Write the trigonometric expression in terms of sine and cosine, and then simplify.

(a)  $\cos t \csc t$

(b)  $\tan^2 x - \sec^2 x$

(c)  $\frac{\sec x}{\csc x}$

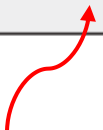
## ACTIVITY 1.1

# 1.1 Simplifying trigonometric expressions

# Proving Trigonometric identities

## GUIDELINES FOR PROVING TRIGONOMETRIC IDENTITIES

1. **Start with one side.** Pick one side of the equation, and write it down. Your goal is to transform it into the other side. It's usually easier to start with the more complicated side.
2. **Use known identities.** Use algebra and the identities you know to change the side you started with. Bring fractional expressions to a common denominator, factor, and use the fundamental identities to simplify expressions.
3. **Convert to sines and cosines.** If you are stuck, you may find it helpful to rewrite all functions in terms of sines and cosines.



Only operations that are **reversible** will necessarily transform an identity into an identity.

Operation of **squaring is not reversible** as we cannot arrive back at the original equation by taking square roots

# Rewriting in terms of Sine and Cosine

Consider the equation  $\cos \theta (\sec \theta - \cos \theta) = \sin^2 \theta$ .

(a) Verify algebraically that the equation is an identity.

The left-hand side looks more complicated, so we start with it and try to transform it into the right-hand side.

$$\begin{aligned}\text{LHS} &= \cos \theta (\sec \theta - \cos \theta) \\ &= \cos \theta \left( \frac{1}{\cos \theta} - \cos \theta \right) && \text{Reciprocal identity} \\ &= 1 - \cos^2 \theta && \text{Expand} \\ &= \sin^2 \theta = \text{RHS} && \text{Pythagorean identity}\end{aligned}$$

EXAMPLE



# Introducing “something extra”

Verify the identity  $\frac{\cos u}{1 - \sin u} = \sec u + \tan u$

**SOLUTION** We start with the left-hand side and multiply the numerator and denominator by  $1 + \sin u$ .

$$\begin{aligned}
 \text{LHS} &= \frac{\cos u}{1 - \sin u} \\
 &= \frac{\cos u}{1 - \sin u} \cdot \frac{1 + \sin u}{1 + \sin u} && \text{Multiply numerator and denominator by } 1 + \sin u \\
 &= \frac{\cos u (1 + \sin u)}{1 - \sin^2 u} && \text{Expand denominator} \\
 &= \frac{\cos u (1 + \sin u)}{\cos^2 u} && \text{Pythagorean identity} \\
 &= \frac{1 + \sin u}{\cos u} && \text{Cancel common factor} \\
 &= \frac{1}{\cos u} + \frac{\sin u}{\cos u} && \text{Separate into two fractions} \\
 &= \sec u + \tan u && \text{Reciprocal identities}
 \end{aligned}$$

EXAMPLE

Verify the identity using the fundamental trigonometric identities.

$$(a) \quad \frac{1 + \sin x}{1 - \sin x} - \frac{1 - \sin x}{1 + \sin x} = 4 \tan x \sec x$$

$$(b) \quad \frac{1 - \cos \alpha}{\sin \alpha} = \frac{\sin \alpha}{1 + \cos \alpha}$$

$$(c) \quad \frac{\sin x - 1}{\sin x + 1} = \frac{-\cos^2 x}{(\sin x + 1)^2}$$

# 1.2a Proving identities

(a)  $\frac{1 + \sin x}{1 - \sin x} - \frac{1 - \sin x}{1 + \sin x} = 4 \tan x \sec x$

SOLUTION

# 1.2b Proving identities

(b)  $\frac{1 - \cos \alpha}{\sin \alpha} = \frac{\sin \alpha}{1 + \cos \alpha}$

SOLUTION

# 1.2c Proving identities

$$(c) \quad \frac{\sin x - 1}{\sin x + 1} = \frac{-\cos^2 x}{(\sin x + 1)^2}$$

SOLUTION

# Trigonometric Substitution

Substitute  $\sin \theta$  for  $x$  in the expression  $\sqrt{1 - x^2}$ , and simplify. Assume that  $0 \leq \theta \leq \pi/2$ .

**SOLUTION** Setting  $x = \sin \theta$ , we have

$$\begin{aligned}\sqrt{1 - x^2} &= \sqrt{1 - \sin^2 \theta} && \text{Substitute } x = \sin \theta \\ &= \sqrt{\cos^2 \theta} && \text{Pythagorean identity} \\ &= \cos \theta && \text{Take square root}\end{aligned}$$

The last equality is true because  $\cos \theta \geq 0$  for the values of  $\theta$  in question.

# Simplifying trigonometric expressions

Make the indicated trigonometric substitution in the given algebraic expression and simplify

Assume that  $0 < \theta < \pi/2$

(a)  $\frac{x}{\sqrt{1-x^2}}, x = \sin \theta$

(b)  $\sqrt{1+x^2}, x = \tan \theta$

(c)  $\sqrt{x^2-1}, x = \sec \theta$

# ACTIVITY 1.3

# 1.3(a) Trigonometric substitution

(a)  $\frac{x}{\sqrt{1-x^2}}, x = \sin \theta$

SOLUTION



# 1.3 (b) Trigonometric substitution

(b)  $\sqrt{1+x^2}, x = \tan \theta$

SOLUTION

# 1.3(c) Trigonometric substitution

(c)  $\sqrt{x^2 - 1}, x = \sec \theta$

SOLUTION

## 2. Addition and Subtraction formulas

# Addition and Subtraction formulas

## ADDITION AND SUBTRACTION FORMULAS

**Formulas for sine:**  $\sin(s + t) = \sin s \cos t + \cos s \sin t$

$$\sin(s - t) = \sin s \cos t - \cos s \sin t$$

**Formulas for cosine:**  $\cos(s + t) = \cos s \cos t - \sin s \sin t$

$$\cos(s - t) = \cos s \cos t + \sin s \sin t$$

**Formulas for tangent:**  $\tan(s + t) = \frac{\tan s + \tan t}{1 - \tan s \tan t}$

$$\tan(s - t) = \frac{\tan s - \tan t}{1 + \tan s \tan t}$$

# Proofs of cofunction identities

## Cofunction Identities

$$\sin\left(\frac{\pi}{2} - x\right) = \cos x \quad \tan\left(\frac{\pi}{2} - x\right) = \cot x \quad \sec\left(\frac{\pi}{2} - x\right) = \csc x$$

$$\cos\left(\frac{\pi}{2} - x\right) = \sin x \quad \cot\left(\frac{\pi}{2} - x\right) = \tan x \quad \csc\left(\frac{\pi}{2} - x\right) = \sec x$$

$$\sin\left(\frac{\pi}{2} - x\right) = \sin\frac{\pi}{2} \times \cos x - \cos\frac{\pi}{2} \times \sin x = 1 \times \cos x - 0 \times \sin x = \cos x$$

$$\cos\left(\frac{\pi}{2} - x\right) = \cos\frac{\pi}{2} \times \cos x + \sin\frac{\pi}{2} \times \sin x = 0 \times \cos x + 1 \times \sin x = \sin x$$

Question: Can you use the subtraction formula for tan to prove the cofunction identity for tan?

$$\tan(s - t) = \frac{\tan s - \tan t}{1 + \tan s \tan t}$$

EXAMPLE

# Using addition and subtraction formulas

Find the exact value of each expression.

(a)  $\cos 75^\circ$       (b)  $\cos \frac{\pi}{12}$

## SOLUTION

(a) Notice that  $75^\circ = 45^\circ + 30^\circ$ . Since we know the exact values of sine and cosine at  $45^\circ$  and  $30^\circ$ , we use the Addition Formula for Cosine to get

$$\begin{aligned}\cos 75^\circ &= \cos(45^\circ + 30^\circ) \\ &= \cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ \\ &= \frac{\sqrt{2}}{2} \frac{\sqrt{3}}{2} - \frac{\sqrt{2}}{2} \frac{1}{2} = \frac{\sqrt{2}\sqrt{3} - \sqrt{2}}{4} = \frac{\sqrt{6} - \sqrt{2}}{4}\end{aligned}$$

(b) Since  $\frac{\pi}{12} = \frac{\pi}{4} - \frac{\pi}{6}$ , the Subtraction Formula for Cosine gives

$$\begin{aligned}\cos \frac{\pi}{12} &= \cos\left(\frac{\pi}{4} - \frac{\pi}{6}\right) \\ &= \cos \frac{\pi}{4} \cos \frac{\pi}{6} + \sin \frac{\pi}{4} \sin \frac{\pi}{6} \\ &= \frac{\sqrt{2}}{2} \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \frac{1}{2} = \frac{\sqrt{6} + \sqrt{2}}{4}\end{aligned}$$

# Using the Addition formula for Sine

Find the exact value of the expression  $\sin 20^\circ \cos 40^\circ + \cos 20^\circ \sin 40^\circ$ .

**SOLUTION** We recognize the expression as the right-hand side of the Addition Formula for Sine with  $s = 20^\circ$  and  $t = 40^\circ$ . So we have

$$\sin 20^\circ \cos 40^\circ + \cos 20^\circ \sin 40^\circ = \sin(20^\circ + 40^\circ) = \sin 60^\circ = \frac{\sqrt{3}}{2}$$

EXAMPLE

# Using addition and subtraction formulas

**Values of Trigonometric Functions** Use an Addition or Subtraction Formula to find the exact value of the expression

(a)  $\sin 15^\circ$

(b)  $\tan 15^\circ$

(c)  $\sin \frac{19\pi}{12}$

# ACTIVITY 2.1



## 2.1a Using addition and subtraction formulas

(a)

SOLUTION

## 2.1b Using addition and subtraction formulas

(b)

**SOLUTION**

## 2.1c Using addition and subtraction formulas

(c)

SOLUTION

# Rewrite expression as Trig function of one number

Use an Addition or Subtraction Formula to write the expression as a trigonometric function of one number, and then find its exact value.

(a)  $\cos 23^\circ \cos 67^\circ - \sin 23^\circ \sin 67^\circ$

(b)  $\sin 35^\circ \cos 25^\circ + \cos 35^\circ \sin 25^\circ$

(c)  $\sin \frac{3\pi}{4} \cos \frac{\pi}{4} - \cos \frac{3\pi}{4} \sin \frac{\pi}{4}$

## ACTIVITY 2.2

## 2.2 Rewrite expression as Trig function of one number

**SOLUTION**

# Expressions with Inverse Trig functions

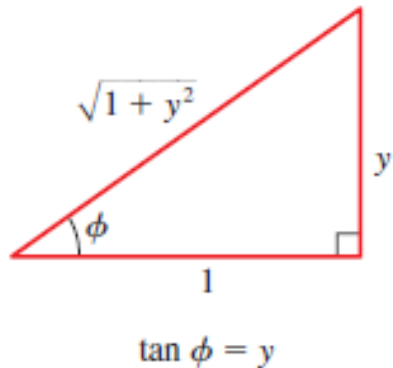
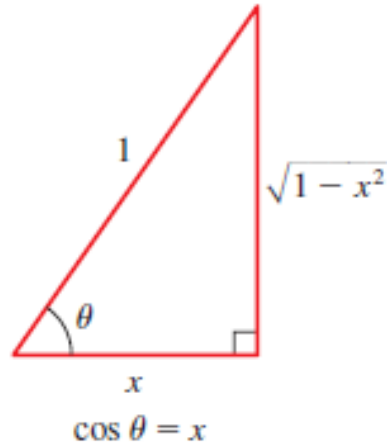


FIGURE 2

## EXAMPLE 6 ■ Simplifying an Expression Involving Inverse Trigonometric Functions

Write  $\sin(\cos^{-1}x + \tan^{-1}y)$  as an algebraic expression in  $x$  and  $y$ , where  $-1 \leq x \leq 1$  and  $y$  is any real number.

**SOLUTION** Let  $\theta = \cos^{-1}x$  and  $\phi = \tan^{-1}y$ . Using the methods of Section 6.4, we sketch triangles with angles  $\theta$  and  $\phi$  such that  $\cos \theta = x$  and  $\tan \phi = y$  (see Figure 2). From the triangles we have

$$\sin \theta = \sqrt{1-x^2} \quad \cos \phi = \frac{1}{\sqrt{1+y^2}} \quad \sin \phi = \frac{y}{\sqrt{1+y^2}}$$

From the Addition Formula for Sine we have

$$\begin{aligned} \sin(\cos^{-1}x + \tan^{-1}y) &= \sin(\theta + \phi) \\ &= \sin \theta \cos \phi + \cos \theta \sin \phi \\ &= \sqrt{1-x^2} \frac{1}{\sqrt{1+y^2}} + x \frac{y}{\sqrt{1+y^2}} \\ &= \frac{1}{\sqrt{1+y^2}} (\sqrt{1-x^2} + xy) \end{aligned}$$

Addition Formula  
for Sine

From triangles

Factor  $\frac{1}{\sqrt{1+y^2}}$

# Expressions with inverse trig functions

Write the given expression in terms of  $x$  and  $y$  only (assume  $x$  and  $y$  are positive).

(a)  $\cos(\sin^{-1}x - \tan^{-1}y)$

(b)  $\tan(\sin^{-1}x + \cos^{-1}y)$

## 2.3a Expressions with inverse trig functions

**SOLUTION**



## 2.3b Expressions with inverse trig functions

**SOLUTION**

# 3. Double-angle and Half-angle Formulas

# What are double angle and half angle formulas?

The **Double-Angle Formulas** allow us to find the values of the trigonometric functions at  $2x$  from their values at  $x$ .

The **Half-Angle Formulas** relate the values of the trigonometric functions at  $\frac{1}{2}x$  to their values at  $x$ .

# Addition and Subtraction formulas

## ADDITION AND SUBTRACTION FORMULAS

**Formulas for sine:**  $\sin(s + t) = \sin s \cos t + \cos s \sin t$

$$\sin(s - t) = \sin s \cos t - \cos s \sin t$$

**Formulas for cosine:**  $\cos(s + t) = \cos s \cos t - \sin s \sin t$

$$\cos(s - t) = \cos s \cos t + \sin s \sin t$$

**Formulas for tangent:**  $\tan(s + t) = \frac{\tan s + \tan t}{1 - \tan s \tan t}$

$$\tan(s - t) = \frac{\tan s - \tan t}{1 + \tan s \tan t}$$

# Double angle formulas

## DOUBLE-ANGLE FORMULAS

**Formula for sine:**  $\sin 2x = 2 \sin x \cos x$

**Formulas for cosine:**  $\cos 2x = \cos^2 x - \sin^2 x$   
 $= 1 - 2 \sin^2 x$   
 $= 2 \cos^2 x - 1$

**Formula for tangent:**  $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$

# Proof of Double-Angle Formulas for Cosine

$$\begin{aligned}\cos 2x &= \cos(x + x) \\ &= \cos x \cos x - \sin x \sin x \\ &= \cos^2 x - \sin^2 x\end{aligned}$$

The second and third formulas for  $\cos 2x$  are obtained from the formula we just proved and the Pythagorean identity. Substituting  $\cos^2 x = 1 - \sin^2 x$  gives

$$\begin{aligned}\cos 2x &= \cos^2 x - \sin^2 x \\ &= (1 - \sin^2 x) - \sin^2 x \\ &= 1 - 2 \sin^2 x\end{aligned}$$

The third formula is obtained in the same way, by substituting  $\sin^2 x = 1 - \cos^2 x$ .

# Using Double angle formulas

If  $\cos x = -\frac{2}{3}$  and  $x$  is in Quadrant II, find  $\cos 2x$  and  $\sin 2x$ .

**SOLUTION** Using one of the Double-Angle Formulas for Cosine, we get

$$\begin{aligned}\cos 2x &= 2 \cos^2 x - 1 \\ &= 2 \left( -\frac{2}{3} \right)^2 - 1 = \frac{8}{9} - 1 = -\frac{1}{9}\end{aligned}$$

To use the formula  $\sin 2x = 2 \sin x \cos x$ , we need to find  $\sin x$  first. We have

$$\sin x = \sqrt{1 - \cos^2 x} = \sqrt{1 - \left( -\frac{2}{3} \right)^2} = \frac{\sqrt{5}}{3}$$

where we have used the positive square root because  $\sin x$  is positive in Quadrant II.  
Thus

$$\begin{aligned}\sin 2x &= 2 \sin x \cos x \\ &= 2 \left( \frac{\sqrt{5}}{3} \right) \left( -\frac{2}{3} \right) = -\frac{4\sqrt{5}}{9}\end{aligned}$$

# Double angle formulas

Find  $\sin 2x$ ,  $\cos 2x$  and  $\tan 2x$  from the given information.

(a)  $\sin x = \frac{5}{13}$ ,  $x$  in Quadrant I

(b)  $\tan x = -\frac{4}{3}$ ,  $x$  in Quadrant II

# ACTIVITY 1.1



# 1.1a Double angle formulas

SOLUTIONS

# 1.1b Double angle formulas

# Half Angle Formulas

Half Angle Formulas allow us to write any trigonometric expression involving even powers of sine and cosine in terms of the first power of cosine only

## FORMULAS FOR LOWERING POWERS

$$\sin^2 x = \frac{1 - \cos 2x}{2} \quad \cos^2 x = \frac{1 + \cos 2x}{2}$$

$$\tan^2 x = \frac{1 - \cos 2x}{1 + \cos 2x}$$

## HALF-ANGLE FORMULAS

$$\sin \frac{u}{2} = \pm \sqrt{\frac{1 - \cos u}{2}} \quad \cos \frac{u}{2} = \pm \sqrt{\frac{1 + \cos u}{2}}$$

$$\tan \frac{u}{2} = \frac{1 - \cos u}{\sin u} = \frac{\sin u}{1 + \cos u}$$

The choice of the + or - sign depends on the quadrant in which  $u/2$  lies.

# Proof of Half-Angle Formulas

**Proof** We substitute  $x = u/2$  in the formulas for lowering powers and take the square root of each side. This gives the first two Half-Angle Formulas. In the case of the Half-Angle Formula for Tangent we get

$$\begin{aligned}\tan \frac{u}{2} &= \pm \sqrt{\frac{1 - \cos u}{1 + \cos u}} \\ &= \pm \sqrt{\left(\frac{1 - \cos u}{1 + \cos u}\right) \left(\frac{1 - \cos u}{1 - \cos u}\right)} && \text{Multiply numerator and denominator by } 1 - \cos u \\ &= \pm \sqrt{\frac{(1 - \cos u)^2}{1 - \cos^2 u}} && \text{Simplify} \\ &= \pm \frac{|1 - \cos u|}{|\sin u|} && \begin{aligned} \sqrt{A^2} &= |A| \\ \text{and } 1 - \cos^2 u &= \sin^2 u \end{aligned}\end{aligned}$$

Now,  $1 - \cos u$  is nonnegative for all values of  $u$ . It is also true that  $\sin u$  and  $\tan(u/2)$  always have the same sign. (Verify this.) It follows that

$$\tan \frac{u}{2} = \frac{1 - \cos u}{\sin u}$$

The other Half-Angle Formula for Tangent is derived from this by multiplying the numerator and denominator by  $1 + \cos u$ . ■

# 1.2 Lowering powers in a trig expression

Use the formulas for lowering powers to rewrite the expression in terms of the first power of cosine

(a)  $\sin^4 x$

(b)  $\cos^4 x$

# ACTIVITY 1.2

# 1.2a Lowering powers in a trig expression

SOLUTION

## 1.2b Lowering powers in a trig expression

SOLUTION

# Using a Half-Angle Formula

Find the exact value of  $\sin 22.5^\circ$ .

**SOLUTION** Since  $22.5^\circ$  is half of  $45^\circ$ , we use the Half-Angle Formula for Sine with  $u = 45^\circ$ . We choose the  $+$  sign because  $22.5^\circ$  is in the first quadrant.

$$\begin{aligned}
 \sin \frac{45^\circ}{2} &= \sqrt{\frac{1 - \cos 45^\circ}{2}} && \text{Half-Angle Formula} \\
 &= \sqrt{\frac{1 - \sqrt{2}/2}{2}} && \cos 45^\circ = \sqrt{2}/2 \\
 &= \sqrt{\frac{2 - \sqrt{2}}{4}} && \text{Common denominator} \\
 &= \frac{1}{2}\sqrt{2 - \sqrt{2}} && \text{Simplify}
 \end{aligned}$$

EXAMPLE



# Half angle formulas

Use an appropriate Half-Angle Formula to find the exact value of the expression.

(a)  $\sin 15^\circ$

(b)  $\tan 15^\circ$

(c)  $\sin 75^\circ$

(d)  $\cos 165^\circ$

# 1.3 Half angle formulas

# Expressions with inverse trig functions

# Expressions with inverse trig functions

Find the exact value of the given expression.

(a)  $\sin \left( 2 \cos^{-1} \left( \frac{7}{25} \right) \right)$

(b)  $\cos \left( 2 \tan^{-1} \left( \frac{12}{5} \right) \right)$

# ACTIVITY 1.4

# SOLUTIONS